

Assignment 2

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```
library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr    1.5.1
## v ggplot2    3.5.1      v tibble     3.2.1
## v lubridate  1.9.3      v tidyr      1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

cohort_data <- read.csv("/Users/claudiagonzalez/Desktop/Data/raw-data/cohort.csv")
#factor all the categorical variables, #0 reference level for all
cohort_data$female <- factor(cohort_data$female, levels = c(0, 1))
cohort_data$cardiac <- factor(cohort_data$cardiac, levels = c(0, 1))
cohort_data$smoke <- factor(cohort_data$smoke, levels = c(0, 1))
cohort_data$age <- as.numeric(as.integer(cohort_data$age)) #changing to numeric vs integer
summary(cohort_data) #summary statistics for cohort data

##   smoke   female      age      cardiac      cost
## 0:4211  0:2226  Min.   :18.00  0:4750  Min.    : 7878
## 1: 789  1:2774  1st Qu.:31.00  1: 250  1st Qu.: 8874
##                Median :44.00                Median : 9143
##                Mean   :43.94                Mean   : 9166
##                3rd Qu.:57.00                3rd Qu.: 9426
##                Max.   :70.00                Max.   :10790

model_cohort <- lm(cost ~ cardiac + age + smoke + female, data = cohort_data)
summary(model_cohort)

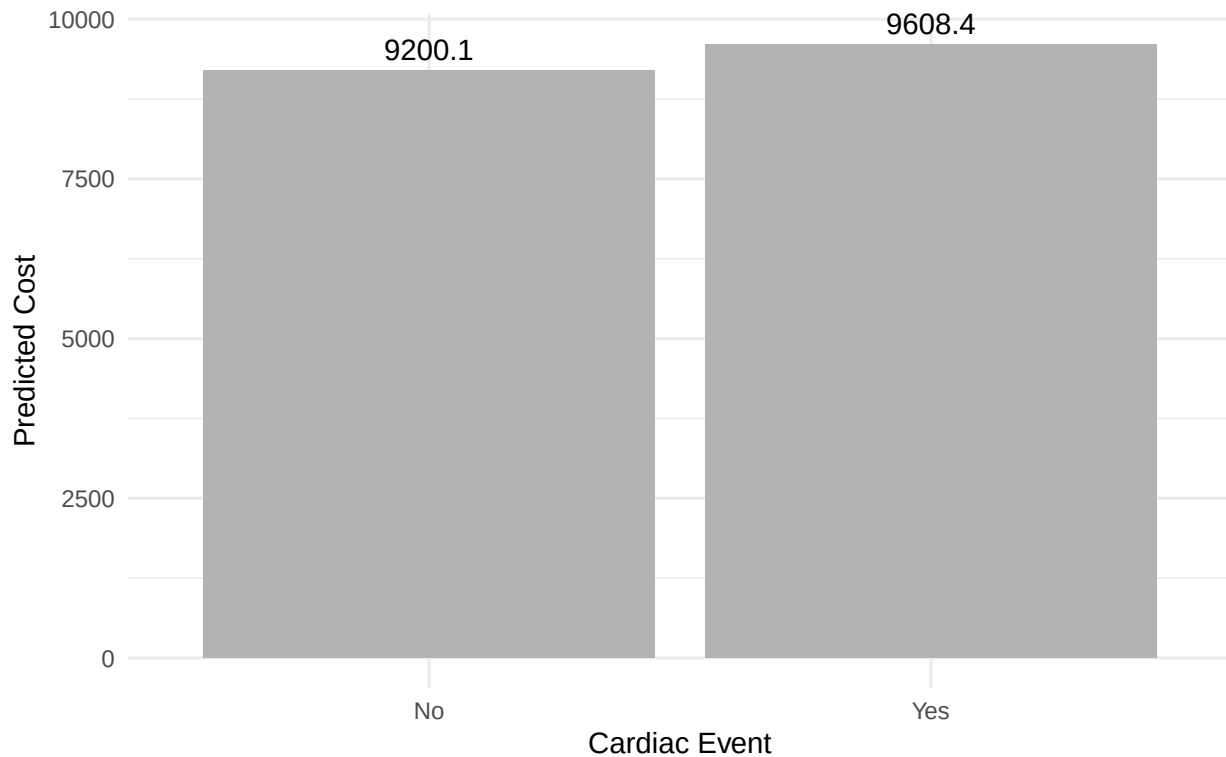
##
## Call:
## lm(formula = cost ~ cardiac + age + smoke + female, data = cohort_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -830.33 -137.76   8.22  140.69  889.72
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 8506.8687     9.8643  862.39  <2e-16 ***
## cardiac1    408.2382     14.2247   28.70  <2e-16 ***
```

```
## age          15.7782      0.1953   80.81   <2e-16 ***
## smoke1       541.9547      8.3271   65.08   <2e-16 ***
## female1     -252.9474      6.0622  -41.73   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 208.3 on 4995 degrees of freedom
## Multiple R-squared:  0.7551, Adjusted R-squared:  0.7549
## F-statistic: 3850 on 4 and 4995 DF,  p-value: < 2.2e-16
#plotting relationship between cardiac event and costs, holding fixed gender, age, and smoking
cohort_subset <- data.frame(
  cardiac = factor(c(0, 1), levels = levels(cohort_data$cardiac)),
  age      = mean(cohort_data$age),
  smoke    = factor(0, levels = levels(cohort_data$smoke)),
  female   = factor(0, levels = levels(cohort_data$female))
)
# predicted costs of a cardiac patient vs. non cardiac patient, with average age, male, non-smoker
cohort_subset$pred_cost <- predict(model_cohort, cohort_subset)

#plot
ggplot(cohort_subset, aes(x = factor(cardiac, labels = c("No", "Yes")), y = pred_cost)) +
  geom_col(fill = "gray70") +
  geom_text(aes(label = round(pred_cost, 1)), vjust = -0.5) +
  labs(
    x = "Cardiac Event",
    y = "Predicted Cost",
    title = "Predicted Cost by Cardiac Status from linear model",
    subtitle = paste0("(at age=", round(mean(cohort_data$age), 1),
                      ", nonsmoker, male)")
  ) +
  theme_minimal()
```

Predicted Cost by Cardiac Status from linear model

(at age=43.9, nonsmoker, male)



```
# Plotting residuals
cohort_data$cardiac <- as.numeric(as.character(cohort_data$cardiac))
cost_resid <- resid(model_cohort)
cardiac_resid <- resid(lm(cardiac ~ age + smoke + female, data = cohort_data))

df_resid <- data.frame(cost_resid, cardiac_resid)
str(df_resid)

## 'data.frame': 5000 obs. of 2 variables:
## $ cost_resid : num 261.73 -233.27 -5.72 56.67 -244.38 ...
## $ cardiac_resid: num 0.0103 0.0131 0.0109 -0.0765 0.0176 ...

ggplot(df_resid, aes(x = cardiac_resid, y = cost_resid)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    x = "Residual cardiac",
    y = "Residual cost",
    title = "Residuals of cost vs. cardiac variable (controlling for age, smoking status, gender)"
  ) +
  theme_minimal()

## `geom_smooth()` using formula = 'y ~ x'
```

Residuals of cost vs. cardiac variable (controlling for age, smoking status,

